

Logical Bilateralism: Proofs, Models and Applications

Session 1 — Foundations: Models, Natural Deduction, and Sequent Calculus

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ESSLLI '26 · Prague

August 3, 2026

Introducing Us

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Course Map

1. Preliminaries: Natural deduction, sequent calculus, models, and some philosophy
2. Signed bilateralist natural deduction and its variations
3. Non-classical frameworks: First Degree Entailment, 2Int, and more
4. Bilateralizing the Curry-Howard correspondence, proofs and refutations, mild and thoroughgoing bilateralism.
5. Recent Developments, Questions to ask, Directions to Pursue

Online Resources

Slides and references are available here:

`https:
//consequently.org/class/2026/bilat-essli/`

Today's Session

My job is to establish our background terminology, and to motivate bilateralism.

- Natural Deduction
- Sequent Calculus
- Models
- Towards Bilateralism

Two routes to validity

An argument from premises P to conclusion C is **valid** if and only if

The Way of Proofs: there is a proof *from* P *to* C . — Write this: ' $P \vdash C$ '

The Way of Models: there is no model in which each member of P *holds* and C *fails*. (Such a model is called a *counterexample* to the argument.) — Write this: ' $P \models C$ '

Soundness: if $P \vdash C$ then $P \models C$

If there is a proof from P to C ,
then this argument has no counterexample.

Typically straightforward to show,
by induction on the structure of the given proof from P to C .

Completeness: if $P \models C$ then $P \vdash C$

If there is no counterexample for the argument from P to C ,
then that argument is backed by some proof.

This is harder to prove.

We must make something from nothing.

Perspectives on Soundness and Completeness

- **Models first:** models provide the *semantics*, while a proof system is judged on the basis of whether it delivers the same consequence relation as the models.
- **Proofs first:** proofs play the semantic role. Models are an idealisation useful as witnesses to counterexamples.
- **Hybrid:** Both proofs and models play a significant semantic role.

Natural Deduction

A proof from $p \rightarrow q$ to $\neg(p \wedge \neg q)$

$$\begin{array}{c}
 \frac{[p \wedge \neg q]^1}{\neg q} \wedge E \qquad \frac{p \rightarrow q \quad \frac{[p \wedge \neg q]^1}{p} \wedge E}{q} \rightarrow E \\
 \hline
 \neg q \quad q \\
 \hline
 \# \\
 \hline
 \neg(p \wedge \neg q) \neg I^1
 \end{array}$$

Rules

$$\begin{array}{c}
 A \qquad \frac{[A]^i \quad \Pi \quad B}{A \rightarrow B} \rightarrow I^i \qquad \frac{\frac{\Pi}{A \rightarrow B} \quad \frac{\Pi'}{A}}{B} \rightarrow E
 \end{array}$$

$$\frac{\frac{\Pi}{A} \quad \frac{\Pi'}{B}}{A \wedge B} \wedge I \qquad \frac{\frac{\Pi}{A \wedge B}}{A} \wedge E \qquad \frac{\frac{\Pi}{A \wedge B}}{B} \wedge E$$

$$\frac{[A]^i \quad \Pi \quad \#}{\neg A} \neg I^i \qquad \frac{\frac{\Pi}{\neg A} \quad \frac{\Pi'}{A}}{\#} \neg E \qquad \frac{\frac{\Pi}{\#}}{A} \#E$$

Rules

$$\begin{array}{c}
 \frac{\Pi}{A} \\
 \hline
 A \vee B
 \end{array} \vee I
 \qquad
 \frac{\Pi}{B} \\
 \hline
 A \vee B
 \end{array} \vee I
 \qquad
 \frac{\Pi \quad [A]^i \quad \Pi' \quad [B]^j \quad \Pi''}{A \vee B \quad C \quad C} \vee E^{i,j}$$

Discharging

Discharging of assumptions is subtle:

- *Contraction*: we can discharge more than one copy of an assumption at a time. (An ‘assumption class’ can have size > 1 .)
- *Weakening*: we can discharge *vacuously*. (An assumption class can have size < 1 .)

$$\begin{array}{c}
 \frac{p \rightarrow (p \rightarrow q) \quad [p]^1}{p \rightarrow q} \rightarrow E \\
 \frac{\frac{p \rightarrow q}{q} \rightarrow I^1 \quad [p]^1}{p \rightarrow q} \rightarrow E
 \end{array}
 \qquad
 \frac{p}{q \rightarrow p} \rightarrow I$$

Normalisation and Harmony

$$\frac{\frac{\frac{[A]^i}{\Pi} B}{A \rightarrow B} \rightarrow I^i \quad \frac{\Pi'}{A} \rightarrow E}{B} \rightsquigarrow \frac{\Pi'}{A \Pi B}$$

$$\frac{\frac{\frac{[A]^i}{\Pi} \#}{\neg A} \neg I^i \quad \frac{\Pi'}{A} \neg E}{\#} \rightsquigarrow \frac{\Pi'}{A \Pi \#}$$

Note: Π' is duplicated if the i assumption class has size > 1
and Π' is deleted if that assumption class has size 0.

Normalisation and Harmony

$$\frac{\frac{\frac{\Pi}{A} \quad \frac{\Pi'}{B}}{A \wedge B} \wedge I}{A} \wedge E \quad \rightsquigarrow \quad \frac{\Pi}{A}$$

$$\frac{\frac{\frac{\Pi}{A}}{A \vee B} \vee I \quad \frac{\frac{[A]^i}{C} \quad \frac{\Pi'}{C}}{C} \quad \frac{\frac{[B]^j}{C} \quad \frac{\Pi''}{C}}{C} \vee E^{i,j}}{C} \quad \rightsquigarrow \quad \frac{\Pi}{A} \quad \frac{\Pi'}{C}$$

Note: Here, a subproof is always deleted: these rules essentially involve weakening.
(Talk to me [GR] if you want to know more about substructural natural deduction.)

Normal Proofs

- These reductions give us some reassurance that the *elimination* rules do not deliver more than we were supplied by the *introduction* rules.
- When you apply these reductions to a proof (together with *permutative* conversions, if the proof uses the disjunction elimination rule) to exhaustion, the result is a *normal* proof.
- A normal proof has the **subformula property**. Every formula in the proof is a subformula of the conclusion or of one of the premises, so these proofs are **analytic** in a strong sense.
- The proof rules for each connective are *separated*. If your argument does not feature a given connective, it has a proof not using that connective's rules.

Classical Logic?

There is a proof from p to $\neg\neg p$ but no proof from $\neg\neg p$ back to p .

$$\frac{\frac{[\neg p]^1 \quad p}{\quad} \neg E}{\frac{\#}{\neg\neg p} \neg I^1}$$

If we add extra rules for negation to get classical logic,
this breaks harmony and normalisation.

Sequent Calculus

Sequents as Bookkeeping

$$\begin{array}{c}
 \frac{[p \wedge \neg q]^1}{\neg q} \wedge E \quad \frac{p \rightarrow q \quad \frac{[p \wedge \neg q]^1}{p} \wedge E}{q} \rightarrow E \\
 \hline
 \frac{\neg q \quad q}{\#} \neg E \\
 \hline
 \neg(p \wedge \neg q) \neg I^1
 \end{array}$$

$$\begin{array}{l}
 p \rightarrow q \succ \neg(p \wedge \neg q) \\
 p \rightarrow q, p \wedge \neg q \succ p \wedge \neg q \succ \neg q p \rightarrow q, p \wedge \neg q \succ q p \wedge \neg q \succ p
 \end{array}$$

Sequents as Bookkeeping

$$\begin{array}{c}
 \frac{[p \wedge \neg q]^1}{\neg q} \wedge E \quad \frac{p \rightarrow q \quad \frac{[p \wedge \neg q]^1}{p} \wedge E}{q} \rightarrow E \\
 \hline
 \frac{\neg q \quad q}{\#} \neg E \\
 \hline
 \frac{\#}{\neg(p \wedge \neg q)} \neg I^1
 \end{array}
 \qquad
 \begin{array}{c}
 \frac{p \succ p}{p \wedge \neg q \succ p} \wedge L \quad \frac{\frac{q \succ q}{q, \neg q \succ} \neg L}{q, p \wedge \neg q \succ} \wedge L \\
 \hline
 \frac{p \rightarrow q, p \wedge \neg q, p \wedge \neg q \succ}{p \rightarrow q, p \wedge \neg q \succ} \rightarrow L \\
 \hline
 \frac{p \rightarrow q, p \wedge \neg q \succ}{p \rightarrow q \succ \neg(p \wedge \neg q)} \text{Contr.} \neg R
 \end{array}$$

The status at each step in a *proof* is given by a *sequent*.

A sequent *derivation* generates a natural deduction *proof*.

Operational Rules

$$\frac{X \succ A \quad X' \succ B}{X, X' \succ A \wedge B} \wedge R \qquad \frac{X, A, B \succ C}{X, A \wedge B \succ C} \wedge L$$

$$\frac{X \succ A}{X \succ A \vee B} \vee R \quad \frac{X \succ B}{X \succ A \vee B} \vee R \qquad \frac{X, A \succ C \quad X', B \succ C}{X, X', A \vee B \succ C} \vee L$$

$$\frac{X, A \succ B}{X \succ A \rightarrow B} \rightarrow R \quad \frac{X \succ A \quad X', B \succ C}{X, X', A \rightarrow B \succ C} \rightarrow L \qquad \frac{X, A \succ}{X \succ \neg A} \neg R \quad \frac{X \succ A}{X, \neg A \succ} \neg L$$

Each operational rule introduces a connective on the right, or on the left.

(Here, A and B are formulas, C is either a formula or is empty, and X and X' are multisets of formulas.)

Structural Rules

$$A \succ A \quad \text{Id} \qquad \frac{X \succ A \quad X', A \succ C}{X, X' \succ C} \text{Cut}$$

$$\frac{X, A, A \succ C}{X, A \succ C} \text{Cont.} \qquad \frac{X \succ C}{X, A \succ C} \text{Wk.L} \qquad \frac{X \succ}{X \succ A} \text{Wk.R}$$

(Here, A and B are formulas, C is either a formula or is empty, and X and X' are multisets of formulas.)

Cut

$$\frac{X \succ A \quad X', A \succ C}{X, X' \succ C} \text{Cut}$$

Cut is the only rule in which a formula (here, A) appears in the premises but not the conclusion.

A derivation without *Cut* has the subformula property: any formula appearing in the derivation appears as a subformula somewhere in the conclusion sequent.

There is a correspondence between *Cut*-free derivations and normal proofs. (1) Any *Cut*-free derivation generates a normal proof. (But some proofs with *Cut* also generate normal proofs.) (2) For any normal proof there is some *Cut*-free proof that generates it.

Symmetric Sequents and Classical Logic

It is *very* easy to see how to modify the sequent calculus to get classical logic.

$$\frac{p \succ p}{p, \neg p \succ} \neg R \qquad \frac{p \succ p}{\succ p, \neg p} \neg L$$

$$\frac{p, \neg p \succ}{p \succ \neg \neg p} \neg L \qquad \frac{\succ p, \neg p}{\neg \neg p \succ p} \neg R$$

$$\frac{\frac{p \succ p}{\neg p, p \succ} \neg R \quad \frac{q \succ q}{\neg q, q \succ} \neg R}{\neg p, \neg q, p \vee q \succ} \vee L$$

$$\frac{\neg p, \neg q, p \vee q \succ}{\neg p \wedge \neg q, p \vee q \succ} \wedge L$$

$$\frac{\neg p \wedge \neg q, p \vee q \succ}{\neg p \wedge \neg q \succ \neg(p \vee q)} \neg R$$

$$\frac{\frac{p \succ p}{\succ \neg p, p} \neg L \quad \frac{q \succ q}{\succ \neg q, q} \neg L}{\succ \neg p, \neg q, p \wedge q} \wedge R$$

$$\frac{\succ \neg p, \neg q, p \wedge q}{\succ \neg p \vee \neg q, p \wedge q} \vee R$$

$$\frac{\succ \neg p \vee \neg q, p \wedge q}{\neg(p \wedge q) \succ \neg p \vee \neg q} \neg L$$

We make sequents symmetric: $X \succ Y$ where X and Y are both multisets.

The rules generalise to the new setting very naturally.

Operational Rules, again

$$\frac{X \succ A, Y \quad X' \succ B, Y'}{X, X' \succ A \wedge B, Y'} \wedge R$$

$$\frac{X, A, B \succ Y}{X, A \wedge B \succ Y} \wedge L$$

$$\frac{X \succ A, Y}{X \succ A \vee B, Y} \vee R$$

$$\frac{X \succ B, Y}{X \succ A \vee B, Y} \vee R$$

$$\frac{X, A \succ Y \quad X', B \succ Y'}{X, X', A \vee B \succ Y, Y'} \vee L$$

$$\frac{X, A \succ B, Y}{X \succ A \rightarrow B, Y} \rightarrow R \quad \frac{X \succ A, Y \quad X', B \succ Y'}{X, X', A \rightarrow B \succ Y, Y'} \rightarrow L$$

$$\frac{X, A \succ Y}{X \succ \neg A, Y} \neg R \quad \frac{X \succ A, Y}{X, \neg A \succ Y} \neg L$$

Each operational rule introduces a connective on the right, or on the left.

(Here, A and B are formulas, and X, X', Y and Y' are multisets of formulas.)

Structural Rules, again

$$A \succ A \quad \text{Id} \qquad \frac{X \succ A, Y \quad X', A \succ Y'}{X, X' \succ Y, Y'} \quad \text{Cut}$$

$$\frac{X, A, A \succ Y}{X, A \succ Y} \quad \text{Cont.L} \qquad \frac{X \succ A, A, Y}{X \succ A, Y} \quad \text{Cont.R} \qquad \frac{X \succ Y}{X, A \succ Y} \quad \text{Wk.L} \qquad \frac{X \succ Y}{X \succ A, Y} \quad \text{Wk.R}$$

(Here, A and B are formulas, and X, X', Y and Y' are multisets of formulas.)

Cut elimination, etc.

The classical sequent calculus has the same good features as the single-conclusion calculus.

Cut can be eliminated from a derivation, to give a derivation with the subformula property.

The sequent calculus is a key tool in many metatheoretic results, like interpolation, consistency proofs, and more.

This results in classical logic, but . . .

$X \succ C$ derivations : natural deduction

$X \succ Y$ derivations : ???

What are classical sequent derivations *about*?

Models

Valuations and validity

A *Boolean valuation* is a map from the language to $\{0, 1\}$ satisfying the usual truth-functional conditions.

We can define validity in the expected way. $X \models A$ iff for every valuation v , whenever $v(B) = 1$ for each $B \in X$ then $v(A) = 1$.

Equivalently, there is no countermodel: no valuation v where $v(X) = 1$ (that's shorthand for $v(B) = 1$ for each $B \in X$) and $v(A) = 0$.

This generalises: $X \models Y$ iff there is no v where $v(X) = 1$ and $v(Y) = 0$.

That is, for any v , if $v(B) = 1$ for each $B \in X$, then for some $C \in Y$, $v(C) = 1$.

It is straightforward to verify, by induction on the construction of the derivation δ , that if δ derives $X \succ Y$ then $X \models Y$.

Positions

Call a pair $[\mathcal{X} : \mathcal{Y}]$ of sets of formulas a *position*.

A position is said to be *available* iff there is no derivation for $X \succ Y$ where X and Y are finite multisets chosen from $\mathcal{X}$ and $\mathcal{Y}$ respectively.

Fact: If $[\mathcal{X} : \mathcal{Y}]$ is available then one of $[\mathcal{X}, A : \mathcal{Y}]$ and $[\mathcal{X} : A, \mathcal{Y}]$ is available.

Why? By *Cut*. If there were a derivation for $X \succ A, Y$ and $X', A \succ Y'$ where X and X' were chosen from $\mathcal{X}$ and Y and Y' were chosen from $\mathcal{Y}$ then $X, X' \succ Y, Y'$ would be derivable and $[\mathcal{X} : \mathcal{Y}]$ would not be available.

Limit Positions

If the *language* $\mathcal{L}$ can be well-ordered, then any available position $[\mathcal{X} : \mathcal{Y}]$ may be extended into a partition $[\mathcal{X}' : \mathcal{Y}']$ of the whole language, where $\mathcal{X} \subseteq \mathcal{X}'$ and $\mathcal{Y} \subseteq \mathcal{Y}'$ and $\mathcal{X}' \cup \mathcal{Y}' = \mathcal{L}$. Call these *limit positions*.

Fact: If $[\mathcal{X} : \mathcal{Y}]$ is a limit position, it has some special properties:

- $\neg A \in \mathcal{X}$ iff $A \in \mathcal{Y}$
- $A \wedge B \in \mathcal{X}$ iff $A, B \in \mathcal{X}$
- $A \vee B \in \mathcal{X}$ iff either $A \in \mathcal{X}$ or $B \in \mathcal{X}$
- $A \rightarrow B \in \mathcal{X}$ iff either $A \in \mathcal{Y}$ or $B \in \mathcal{X}$

Why? Use the operational rules, the fact that $[\mathcal{X} : \mathcal{Y}]$ is available, liberally using the fact that $A \in \mathcal{Y}$ iff $A \notin \mathcal{X}$.

This makes limit positions act like *valuations*.

Completeness, sequent style

Completeness is easy now:

If $X \not\vdash Y$ then the position $[X : Y]$ is available (fudging the set/multiset distinction).

Extend it to a limit position $[\mathcal{X} : \mathcal{Y}]$.

This determines a Boolean valuation that assigns 1 to every member of $\mathcal{X}$ (and so, every member of X) and 0 to every member of $\mathcal{Y}$ (and so, to every member of Y).

So, $X \not\vdash Y$.

How about single-conclusion sequents?

Boolean valuations overgenerate with respect to intuitionistic validity. Let's see what availability and limit positions look like for single-conclusion sequents.

$[\mathcal{X} : \mathcal{Y}]$ is *intuitionist available* (i-available) iff there is no derivation for $X \succ C$ where X is a finite multiset chosen from $\mathcal{X}$ and C is either empty or in $\mathcal{Y}$.

Fact: If $[\mathcal{X} : \mathcal{Y}]$ is i-available then one of $[\mathcal{X}, A : \mathcal{Y}]$ and $[\mathcal{X} : A, \mathcal{Y}]$ is too.

Why? By *Cut*. If there were a derivation for $X \succ A$ and $X', A \succ C$ where X and X' were chosen from $\mathcal{X}$ and C is chosen from $\mathcal{Y}$ then $X, X' \succ C$ would be derivable and $[\mathcal{X} : \mathcal{Y}]$ would not be i-available.

Limit positions for single-conclusion sequents

If $[\mathcal{X} : \mathcal{Y}]$ is an *intuitionist* limit position, some of our special properties fail.

Neither $\vdash p$ nor $\vdash \neg p$, so $[\vdash p, \neg p]$ is i-available. This can extend to a limit position where p and $\neg p$ are both on the right side.

Neither $p \vee q \vdash p$ nor $p \vee q \vdash q$, so $[p \vee q : p, q]$ is i-available. Any limit position extending this will have $p \vee q$ but neither p nor q on the left.

Neither $\vdash p$ nor $\vdash p \rightarrow q$, so $[\vdash p, p \rightarrow q]$ is i-available. This can be extended to a limit position where p and $p \rightarrow q$ are both on the right side.

Of the classical clauses for the connectives, we can only say that in any limit position $[\mathcal{X} : \mathcal{Y}]$ we have $A \wedge B \in \mathcal{X}$ iff $A, B \in \mathcal{X}$.

The clauses for negation, disjunction and the conditional are more complex.

Intuitionist limit position ordering

$[\mathcal{X} : \mathcal{Y}] \sqsubseteq [\mathcal{X}' : \mathcal{Y}']$ when $\mathcal{X} \subseteq \mathcal{X}'$. $[\mathcal{X}' : \mathcal{Y}']$ is a *strengthening* of $[\mathcal{X} : \mathcal{Y}]$.

If $A \rightarrow B \in \mathcal{X}$, then for any i-available $[\mathcal{X}' : \mathcal{Y}']$ strengthening $[\mathcal{X} : \mathcal{Y}]$ where $A \in \mathcal{X}'$ we have $B \in \mathcal{X}'$ too, since $A \rightarrow B, A \vdash B$ (by $\rightarrow E$)

On the other hand, if $A \rightarrow B \in \mathcal{Y}$, then $[\mathcal{X}, A : B]$ must be i-available (since otherwise, by $\rightarrow I$, $[\mathcal{X} : A \rightarrow B]$ would not be i-available), and so, it can be extended to a limit position $[\mathcal{X}' : \mathcal{Y}']$ where $A \in \mathcal{X}'$ and $B \in \mathcal{Y}'$ and $[\mathcal{X} : \mathcal{Y}] \sqsubseteq [\mathcal{X}' : \mathcal{Y}']$.

So: for any i-available $[\mathcal{X} : \mathcal{Y}]$, $A \rightarrow B \in \mathcal{X}$ iff for each strengthening $[\mathcal{X}' : \mathcal{Y}']$ of $[\mathcal{X} : \mathcal{Y}]$ if $A \in \mathcal{X}'$ then $B \in \mathcal{X}'$ too.

Connective behaviour in intuitionist limit positions

Similarly, for any intuitionist available $[\mathcal{X} : \mathcal{Y}]$, $\neg A \in \mathcal{X}$ iff for each strengthening $[\mathcal{X}' : \mathcal{Y}']$ of $[\mathcal{X} : \mathcal{Y}]$, $A \in \mathcal{Y}'$.

For disjunction, we have $A \vee B \in \mathcal{X}$ iff any strengthening of $[\mathcal{X} : \mathcal{Y}]$ has some further strengthening $[\mathcal{X}'' : \mathcal{Y}'']$ where $A \in \mathcal{X}''$ or $B \in \mathcal{X}''$.

Why? Suppose $[\mathcal{X}' : \mathcal{Y}']$ is a strengthening. Since $A \vee B \in \mathcal{X} \subseteq \mathcal{X}'$, one of $[\mathcal{X}', A :]$ and $[\mathcal{X}', B :]$ must be i-available—since if they weren't, $[\mathcal{X}', A \vee B :]$ would not be i-available either (by $\vee$ L)—so the available one can be extended to a limit position with A or with B on the left.

Conversely, if $A \vee B \in \mathcal{Y}$, we want some strengthening of $[\mathcal{X} : \mathcal{Y}]$ which has *no* strengthening with either A or B on the left. Start with $[\mathcal{X} : A, B]$. This must be i-available, since neither A nor B can be proved from anything in $\mathcal{X}$. This can be extended to a limit position, strengthening $[\mathcal{X} : \mathcal{Y}]$ that has *no* strengthening with either A or B on the left.

Intuitionist limit positions form a Beth model

A set P of points, ordered by $\sqsubseteq$, with a support relation $\Vdash$ on atoms satisfying:

- If $x \Vdash p$ and $x \sqsubseteq x'$ then $x' \Vdash p$ too. (*Heredity*)

We extend $\Vdash$ to $\mathcal{L}$ as follows:

- $x \Vdash A \wedge B$ iff $x \Vdash A$ and $x \Vdash B$.
- $x \Vdash A \vee B$ iff $(\forall x' \sqsupseteq x)(\exists x'' \sqsupseteq x')$ where either $x'' \Vdash A$ or $x'' \Vdash B$.
- $x \Vdash A \rightarrow B$ iff $(\forall x' \sqsupseteq x)$, if $x' \Vdash A$ then $x' \Vdash B$.
- $x \Vdash \neg A$ iff $(\forall x' \sqsupseteq x)$, $x' \not\Vdash A$.

A Beth model forms a *counterexample* to an argument if it contains a point that supports the premises and fails to support the conclusion.

Soundness and completeness for intuitionist proofs and Beth frames is now *immediate*.

Kripke frames

We could do this all for *Kripke* frames (with their simpler disjunction clause) by tweaking the definition of availability.

A position $[\mathcal{X} : \mathcal{Y}]$ is *Kripke-available* if there is no derivation of $X \succ B$ where X is a multiset chosen from $\mathcal{X}$ and B is a *disjunction* of members of $\mathcal{Y}$.

For details, see Greg Restall “Truth Values and Proof Theory” *Studia Logica* (2009).

Towards Bilateralism

Models

Boolean valuations exhibit a clear true/false symmetry.

Beth frames and Kripke frames treat truth and falsity asymmetrically.

Truth is preserved over $\sqsubseteq$, while untruth is not preserved.

Falsity is modelled as *modalised* untruth.

This seems to be *detachable* from constructive intuitions. Why not treat truth and falsity on a par, as preserved under $\sqsubseteq$?

Natural Deduction and the Sequent Calculus

Single conclusion sequent calculus has a left/right asymmetry, while the classical sequent calculus is left/right symmetric.

Natural deduction is *clearly* premise/conclusion asymmetric. Proofs have one conclusion (or none), and any number of premises.

Is there a properly *symmetric* form of *proof* for classical logic (other than the sequent calculus)?

The issues of proofs and models are related: *positions* show how the left/right or premise/conclusion distinction in proofs is connected to the polarity between true/false in models.

Summary

- We introduced natural deduction proofs, harmony for the operational rules, and normalisation.
- We introduced the single-conclusion sequent calculus, motivating sequents as explicit bookkeeping for natural deduction proofs. We showed how sequent derivations describe natural deduction proofs, and showed how *Cut*-free derivations are related to normal proofs.
- We introduced the multiple-conclusion classical sequent calculus, showing how it's a natural extension of the single-conclusion system.
- We showed how models appropriate for classical and intuitionist logic arise naturally from a limit construction for positions.
- We raised the question about the asymmetry between truth and falsity or premise and conclusion in models and proofs.

Further Reading

- Dag Prawitz, *Natural Deduction*, Almqvist and Wiskell, 1965.
- Gerhard Gentzen, “Untersuchungen über das logische Schliessen”, *Mathematische Zeitschrift* 39 pp. 176–210, pp. 405–31, 1935, translated as “Investigations into Logical Deduction” in *The Collected Papers of Gerhard Gentzen*, edited by M. E. Szabo, North-Holland, 1969.
- Greg Restall, “Truth Values and Proof Theory,” *Studia Logica*, 92 pp. 241–264, 2009.
- Greg Restall, *Proofs and Models in Philosophical Logic*, Cambridge University Press, 2023.

Tomorrow

Signed bilateralist natural deduction and its variations.

Any Questions?